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Palaeomagnetic Results from Palaeozoic Sediments of Northern Australia

G. R. Luck*

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Summary

A palaeomagnetic study was made of 52 samples from an Upper Devonian formation and 97 samples from three Cambrian formations. In most cases, the application of demagnetization procedures failed to reveal other than a recent magnetization, presumably acquired by chemical means during weathering processes. Seventeen samples, four of which had been collected a decade earlier, appeared to yield the direction of original magnetization of the Middle Cambrian Hudson Formation. Assigning unit weight to sample mean results, the palaeomagnetic pole position for the period of deposition is calculated to be 19° *E*, 18° N with $A_{95} = 13°$. The data for the Jinduckin Formation have been re-assessed using sample results, and these give a slightly revised pole position for the Lower Ordovician at 25° E, 13° S with $A_{95} = 11°$.

1. Introduction

The first results from a recent palaeomagnetic survey of the Palaeozoic of northern Australia indicated the widespread occurrence of a sometimes dominant secondary magnetization (Luck 1970; McElhinny & Luck 1970). The importance of this effect was revealed during measurements on 149 samples from four other sedimentary formations collected in the same general region (Table 1). The natural remanent magnetization (NRM) of most of these samples was directed along the Earth's present field.

A similarly directed secondary component, presumably representing a viscous remanent magnetization (VRM), had been readily removed from most samples of the Jinduckin Formation by the application of partial demagnetization techniques (Luck 1970). In contrast, the present paper reports that recently acquired magnetization could not be cleaned from the Blatchford Formation, the Hart Spring Sandstone, the Cockatoo Formation or from most samples of the Hudson Formation. This hard, secondary component was taken to represent a chemical remanent magnetization (CRM) imposed by weathering processes. In favourable circumstances, the effect might be neutralized by a chemical method such as devised by Collinson (1965) for separating the red pigment from the specularite of red beds. This technique may be regarded as a means of partial demagnetization (Collinson 1967) if the secondary magnetic component resides in the pigment.

Disc-shaped (3·5 cm × 1·0 cm) and cylindrical (2·8 cm × 2·8 cm) specimens were cut from most samples. The discs were investigated using an astatic magnetometer

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Table 1

Formations sampled in northern Australia

Period	Formation	No. of Samples
	Cockatoo Formation	
Upper Devonian	Abney Sandstone	25
	Kellys Knob Sandstone	12
	Ragged Range Conglomerate	15
Lower Ordovician	Jinduckin Formation†	28
Middle Cambrian	Hudson Formation	48*
	Hart Spring Sandstone	27
	Montejinni Limestone‡†	18
Lower Cambrian	Blatchford Formation	22
	Antrim Plateau Volcanics†	99

* Total includes nine samples collected by Irving (see text)

† Results from these formations have been reported by Luck (1970) and McElhinny & Luck (1970).

‡ Stratigraphic relationship to the other Middle Cambrian formations uncertain.

(Collinson *et al.* 1957), the others being measured with a spinner magnetometer (P.A.R. model SM1). The NRM of at least one specimen per sample was determined before applying alternating field (a.f.) and progressive thermal demagnetization techniques. The apparatus for magnetic cleaning was similar to the one described by McElhinny (1966), that for thermal cleaning being described by McElhinny, Luck & Edwards (1971).

Representative discs from all four formations were subjected to unsuccessful attempts at chemical demagnetization, as previously applied to specimens from the Jinduckin Formation. As before, the magnetic intensities were eventually reduced to immeasurable values without any accompanying changes in the directions.

2. The Southern Bonaparte Gulf Basin

2.1 *The Blatchford Formation*

The Blatchford Formation appears as a narrow, 30 km strip of outcrop along the western edge of the Ragged Range in Western Australia (Fig. 1). It is more than 130 m thick and consists of grey, green and brown dolomitic siltstone—interbedded with fossiliferous glauconite and dolomite—which is overlain by cross-bedded, fine-grained, brown, dolomitic quartz sandstone. Fossil-dated as Lower Cambrian, the formation disconformably overlies the Antrim Plateau Volcanics and is overlain, with a marked angular unconformity, by the Ragged Range Conglomerate Member of the Upper Devonian Cockatoo Formation. It dips gently eastwards at angles up to a maximum of 25° (Kaulback & Veevers 1970).

The formation was sampled at two localities: at one, eight samples were obtained from two sites atop a sharp, exposed ridge (Locality 1, Fig. 1), while at the second (Locality 2), some 8 km further north-east, 14 samples were collected from seven sites in the flank of the range. A total of 33 specimens was eventually measured.

Samples from the first two sites proved to be randomly magnetized, the remanent magnetization being of an isothermal nature. Thermal treatment, with temperatures

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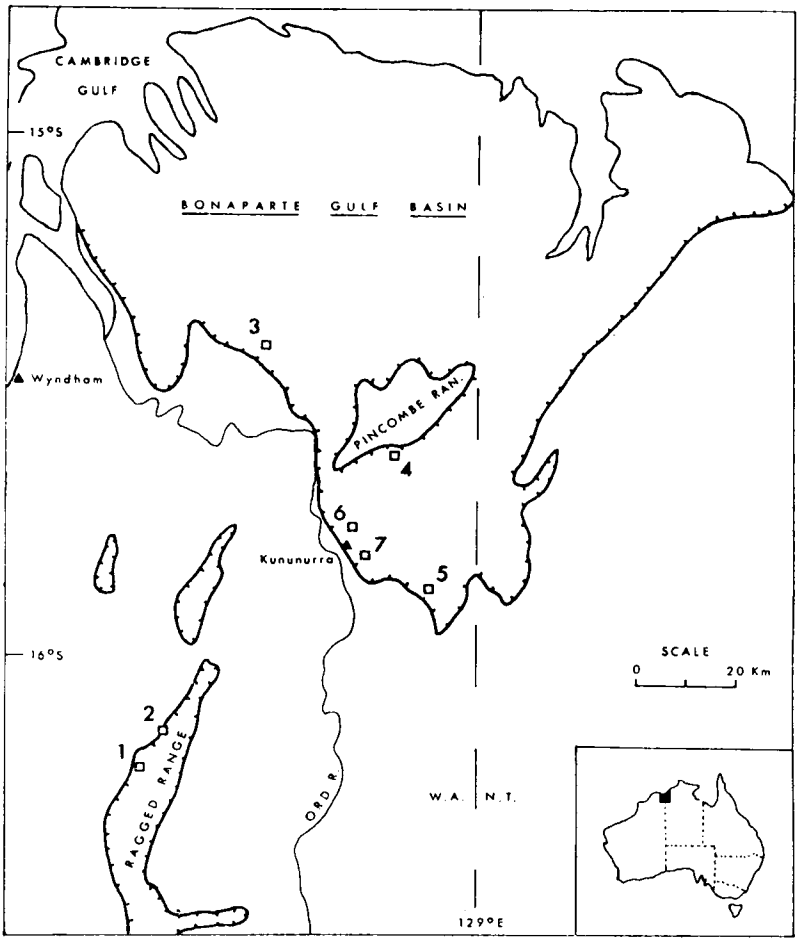


FIG. 1. The southern Bonaparte Gulf Basin and outliers. Sketch map indicating sampling localities.

up to 650° C, failed to affect this IRM. With the application of a.f. methods, specimens revealed a stable magnetization along the direction of the present Earth's field. The NRM directions of samples from the seven other sites were not random and tended strongly towards that of the present geomagnetic field. This proved to be a viscous component that was removed by thermal treatment, specimens reverting to random magnetic directions. However, magnetic cleaning quickly removed both the VRM and the IRM, specimens showing a stable magnetization in the direction of the present Earth's field.

It therefore appears that the Blatchford Formation has been chemically remagnetized in recent times, but that superimposed upon the CRM is an IRM resulting from repeated lightning strikes on the very exposed outcrop. The NRM at Locality 2 is further complicated by the presence of a VRM.

2.2 *The Hart Spring Sandstone*

The Hart Spring Sandstone is found in the Bonaparte Gulf Basin and associated outliers as a member of a well-dated conformable sequence of Middle Cambrian to Lower Ordovician marine sediments. It has a total thickness of about 400 m and is

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comprised of fine to medium-grained red, micaceous quartz sandstone. The red colour emanates from primary haematite, coating the original grains, and secondary haematite, which cements grains enlarged with quartz overgrowths (Kaulback & Veevers 1970). Where exposed, the formation commonly dips to the north-east at about 5°, much steeper dips being observed in the neighbourhood of local faults.

Twenty-seven samples were collected from 11 sites (Locality 3, Fig. 1) and a total of 36 specimens was measured.

The NRM of nearly every sample was in the direction of the Earth's present field and thermal cleaning invariably failed to remove the effects of this recent remagnetization. Two ridge-top sites produced random NRM directions that were unaffected by thermal treatment and had presumably been induced by lightning. Magnetic washing failed to reveal any stable magnetization other than a recent CRM.

2.3 *The Cockatoo Formation*

This well-dated Upper Devonian formation unconformably overlies the Cambrian rocks of the southern Bonaparte Gulf Basin. It is at least 2000 m thick and is comprised of eight members, three of which were sampled.

The Ragged Range Conglomerate Member forms the base of the Cockatoo Formation and consists of a wedge, at least 300 m thick, of boulder conglomerate and red, lithic quartz sandstone. Its most extensive outcrop, to the south of the Bonaparte Gulf Basin proper, constitutes the bulk of the Ragged Range, where the thickness and proportion of conglomerate and sandstone vary considerably. (Veevers & Roberts 1968).

Six samples were collected from three sites just south of the Precambrian Pincombe Inlier (Locality 4, Fig. 1) and nine samples were collected 80 km further south-west, from three sites in the Ragged Range (Locality 2). A total of 32 specimens was measured.

The nine samples from the south were collected near the summit of the range, immediately above the second sampling locality in the Blatchford Formation, and they too were found to exhibit IRM characteristics. Thermal treatment hardly affected their directions of magnetization, while magnetic cleaning tended only to reveal a recently acquired CRM.

The other six samples (from Locality 4) had NRM directions along the present Earth's field. This direction was generally unaffected by partial demagnetization techniques although some specimens disclosed an almost horizontal direction of magnetization after being heated to 550 °C.

The Kellys Knob Sandstone Member is a tabular body, at least 300 m thick, lying just above the basal conglomerate. Except for increasing amounts of pebbles and boulders near the margin of the basin, it is a remarkably uniform, cross-bedded, yellow to white quartz sandstone. (Veevers & Roberts 1968). It does, however, incorporate some red sandstone where 12 samples were collected from five sites (Locality 5, Fig. 1). Thirty-two specimens were measured.

Whilst one site was found to have been recently chemically remagnetized, the other sites possessed NRMs that were randomly directed. Thermal treatment of the latter four sites appeared to remove some secondary components but the resulting directions of magnetization were seemingly meaningless. It is evident that lightning had affected these four sites because isothermal components could be removed by magnetic washing to leave a stable magnetization along the present Earth's field.

Because thermal treatment had affected the magnetization of many specimens, an attempt was made to remove both the IRM and CRM by magnetically cleaning a specimen in a peak alternating field of 400 Oe and then heating it to 520 °C. However, it was found that after a specimen had been magnetically cleaned its magnetism was virtually unaffected by thermal cleaning.

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The Abney Sandstone Member is a sheet, 100 m thick, of thin-bedded red to brown quartz sandstone. Fourteen samples were collected from six sites at Locality 6, Fig. 1, and 11 samples from four sites located 8 km to the south-east (Locality 7).

NRM measurements suggested that all samples had been chemically remagnetized in the direction of the present geomagnetic field. This CRM was not appreciably affected by magnetic, thermal or chemical cleaning methods. The total number of specimens measured was 60.

2.4 *Discussion*

The results recorded in the preceding sections are most likely attributable to the widespread action of recent chemical remagnetization—unless the four formations were originally magnetized in a direction very similar to that of the Earth's present field. This latter possibility may be disregarded for the following reasons.

1. A comparison with the Cambrian data previously reported (Luck 1970; McElhinny & Luck 1970) suggests that an original magnetization for the Blatchford Formation and Hart Spring Sandstone would have been considerably removed from that of the present geomagnetic field.

2. The Cockatoo Formation, of Frasnian Age, is expected to produce a palaeomagnetic pole position close to that of the Upper Devonian Lochiel Formation, or, perhaps, that of the Lower Devonian Ainslie Volcanics (Luck 1972, in preparation). An original magnetization would be inclined downwards at 25° in the former case or be about horizontal in the latter case, whereas the Earth's present and axial dipolar fields in northern Australia are inclined upwards at 45° and 30° respectively.

3. All directions with respect to the horizontal correspond to the present geomagnetic field despite bedding dips as great as 25°.

It is concluded that all four formations have been chemically remagnetized in their present positions.

3. The Hardman Basin

The Hardman Basin is centred about 200 km south of Kununurra where the fossil-dated Negri Group of Middle Cambrian sediments has been downfolded into the Antrim Pleau Volcanics. The Negri Group, with a thickness of perhaps 800 m, is unconformably overlain by the Devonian Elder Sandstone—a very friable, white sandstone possibly 400 m thick.

Most of the Negri Group is comprised of limestones and poorly exposed red and grey shales. The youngest member, the Hudson Formation, appeared to contain the only material in the basin warranting a palaeomagnetic collection.

3.1 *The Hudson Formation*

This formation, formerly known as the Elder Mountain Sandstone (e.g. Irving & Green 1958), has a total thickness of about 150 m. It consists primarily of thin to medium beds of brown, micaceous sandstone (usually cross-bedded and ripple marked) and containing 5–20 cm thick iron beds that approach ore grade. About 20 per cent of the formation is comprised of grey or red, laminated, micaceous siltstone and shale; a thin bed of grey, copper-stained limestone is also present (Dow *et al.* 1964).

3.2 *Sampling*

The localities at which collections were made are shown in Fig. 2. Sites A–O were sampled from a 30 m vertical section that is exposed by a fairly steep creek entering the Ord River. Twenty kilometres farther north-east (i.e. downstream), sites

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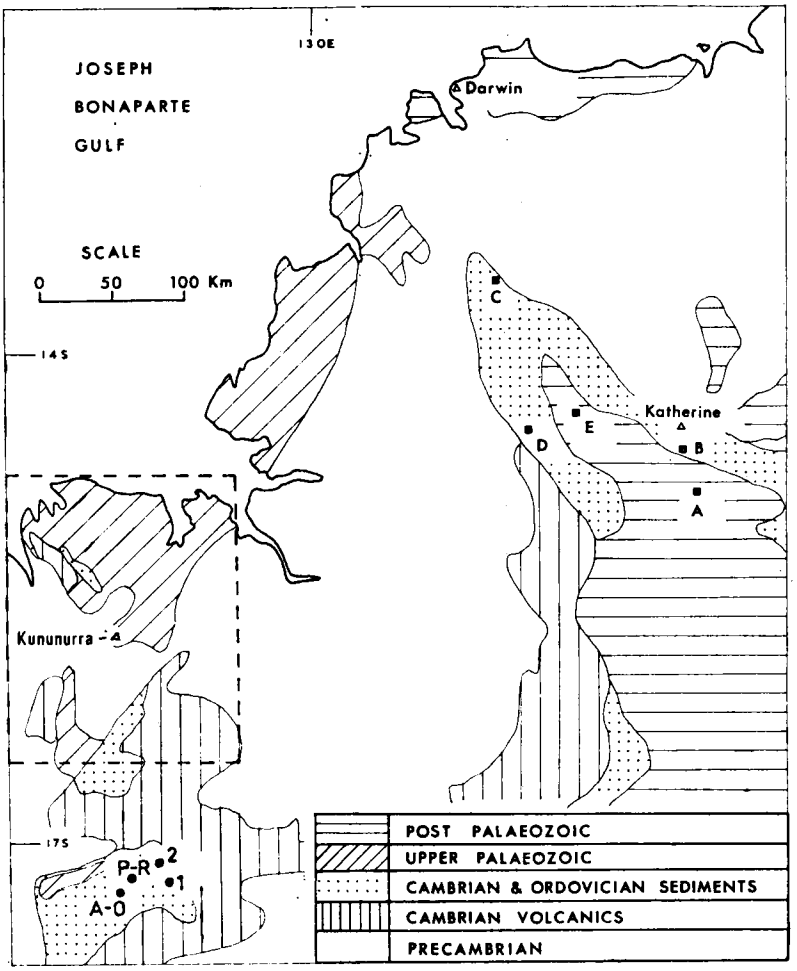


FIG. 2. Geological sketch map of northern Australia indicating the widespread sampling of Palaeozoic sediments. Sampling localities in the Hudson Formation are represented by circles. Also shown are sites in the Jinduckin Formation (Luck 1970) and the area covered by Fig. 1.

P and Q were collected in the bed of the Ord River—from presumably stratigraphically lower horizons. Site *R* was located 4 km east of P and Q at a rubbly hillside exposure, where the beds are tilted at twice the commonly encountered angle of 7°.

Irving (1958) had previously collected nine samples of the so-called Elder Mountain Sandstone from two sites. Site 2 was situated at the Negri River, about 20 km north-east of Site *R* and 20 km north of his Site 1.

Three different sizes of specimens were cut from most samples: discs (3·5 cm × 1·0 cm), small cylinders (2·2 cm × 2·2 cm) and large cylinders (2·8 cm × 2·8 cm); precise discs (3·5 cm × 0·7 cm) had been machined from Irving's samples. Measurements were made on a total of 167 specimens from the 48 samples.

3.3 *Laboratory measurements Results*

Throughout this section a distinction is made between the nine samples of Irving and those collected more recently. This is to emphasize the differences between the two collections in behaviour and results.

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Irving had found that his specimens could be classified into three types according to their NRM directions, which were either (1) close to that of the Earth's present field; or (2) grouped about a direction some 100° from (1); or (3) strung along a portion of the ' great circle ' between (1) and (2).

Irving naturally assumed that specimens of the second type had probably retained their original magnetization and, in calculating a palaeomagnetic pole position for the formation, he accordingly accepted only the data from the three samples whose constituent specimens were exclusively of this type.

The NRMs of all his specimens have been re-measured and subjected to the statistical criteria of Watson (1956) and Vincenz & Bruckshaw (1960), who tabulated values for the vector resultant ' R ' which are exceeded in only 5 per cent of cases by a set of randomly directed vectors. Seven of the samples were found to possess non-random directions of magnetization that, for no apparent reason, were rather different from those reported by Irving, as shown in Fig. 3(a). (Since the specimens had been stored on their edges, a VRM acquired during the past 12 years would have had a random effect). Most of the NRM directions of the other samples were clustered near to that of the present geomagnetic field (Fig. 3 (b)),with some stringing towards the cleaned directions shown in Fig. 6.

Measurements on Irving's samples and on sites A–C were made using the appropriate astatic magnetometer, most of the remaining work being carried out with the spinner magnetometer. The cleaned results that may represent a primary magnetization are listed in Table 2, the rejected data being summarized, for comparison, in Table 3.

3.4 *The assessment of data from sedimentary formations*

When calculating palaeomagnetic pole positions for volcanics, it is normal practice to attach unit weight to the mean results for sites. However, the analysis of data from

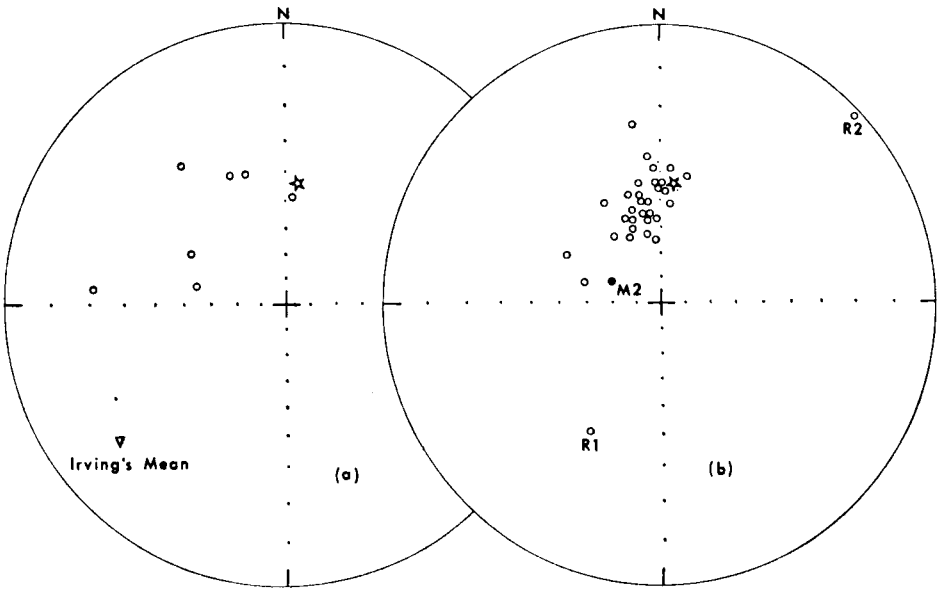


FIG. 3. Hudson Formation—NRM directions. (*a*) The mean directions from the samples collected by Irving and remeasured by the author. The mean direction of NRM reported by Irving is shown as a triangle. (The star represents the present geomagnetic field direction). (*b*) The mean directions from samples recently collected.

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Table 2

Hudson Formation: accepted sample mean directions after cleaning

Sample	n	D	I	R	k	Pole	
A1	3	262	−58	2·70	6·6	00° E	05° N
A2	3	185	−81	2·65	5·7	311° E	00° N
E2	4	278	−27	3·91	34	27° E	11° N
H1	4	285	−55	3·94	49	07° E	22° N
K1	2	293	−30	2·00	220	29° E	26° N
N1	4	305	−37	3·97	99	26° E	38° N
N2	4	316	−35	3·98	150	29° E	48° N
O1	2	308	−37	1·95	21	27° E	41° N
O2	4	292	+06	3·68	9·3	49° E	20° N
P1	2	282	−46	1·99	97	15° E	18° N
Q2	3	306	−45	2·85	14	18° E	39° N
R1 af	3	262	−46	2·95	38	11° E	01° N
CE1	4	268	+10	3·62	7·9	43° E	04° S
CE6	4	282	+19	3·73	11	52° E	08° N
CE7	4	282	−20	3·84	19	32° E	14° N
CE8	2	258	−44	2·00	340	11° E	03° S
CE9	4	241	−63	3·71	10	348° E	07° S

n = Number of specimens measured.

D = Declination in degrees, east of north.

I = Inclination in degrees, positive downwards.

R = Resultant of n unit vectors.

k = Fisher s precision parameter.

α_{95}, A_{95} = Semi-angles of cones of 95 per cent confidence for mean direction and mean pole respectively.

R1 af = Cleaned in peak field of 500 Oe; all other samples cleaned thermally at temperatures in the range 450° C−600° C.

sedimentary formations is less straightforward, and different workers have variously used results from either specimens, samples or sites—with some workers changing their systems according to the particular problem.

Although site mean data were used in the original analysis of results from the Jinduckin Formation, it now seems that a more appropriate method would have been to assign unit weight to samples. This conclusion proceeds partly from the observation that for the Jinduckin and Hudson Formations, as for many other sediments, the within-site dispersions of directions of magnetization are comparable to the between-site dispersions. (The fact that a sample mean direction may represent a dipolar geomagnetic field is illustrated by the presence of a reversal within site C1 of the Jinduckin Formation). In addition, it was found that many sites in the Hudson Formation contained both recently remagnetized and non-remagnetized samples.

In most cases, regardless of the above stated preference for sample means, the calculated position of a palaeomagnetic pole would not be greatly affected by which of the two means is accorded unit weight. This is because random errors, arising from the methods of collection and measurement, are largely eliminated when averaging the results from either many samples or many sites; any systematic errors would bias both types of calculation to a similar extent. Nevertheless, it is noted that the degree of confidence in a palaeomagnetic pole position, as characterized by the value of A_{95}, will usually appear greater (for a particular formation) where unit weight is given to samples.

The data from the Jinduckin Formation have been re-assessed following the measurement of several more specimens and using the means of samples rather than

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sites. The revised calculations have incorporated the two results from site C1, and the amended pole position is 25° E, 13° S with $A_{95} = 11°$ (compared with 29° E, 16° S and $A_{95} = 17°$). The original and revised results are summarized in Table 3.

3.5 *Discussion of results from the Hudson Formation*

The streaked directions of Fig. 3 illustrate the presence in most samples of a secondary magnetization along the Earth's present field. This recently acquired magnetism is in many cases a combination of both a VRM and a CRM, the former being readily removed by thermal treatment, as indicated by an initial rapid decrease in intensity. The ratio of the intensities of secondary to primary CRM varied between samples, as depicted by the different levels of plateaus of the curves in Fig. 4. A similar observation concerning Permian red beds was made by Farrell & May (1969) who also observed the necessity for applying some criterion of selection (usually based on the behaviour of the samples during demagnetization procedures) when assessing the results from remagnetized rocks. For example, they accepted the data from only 32 out of 308 specimens from one locality, although the criterion adopted

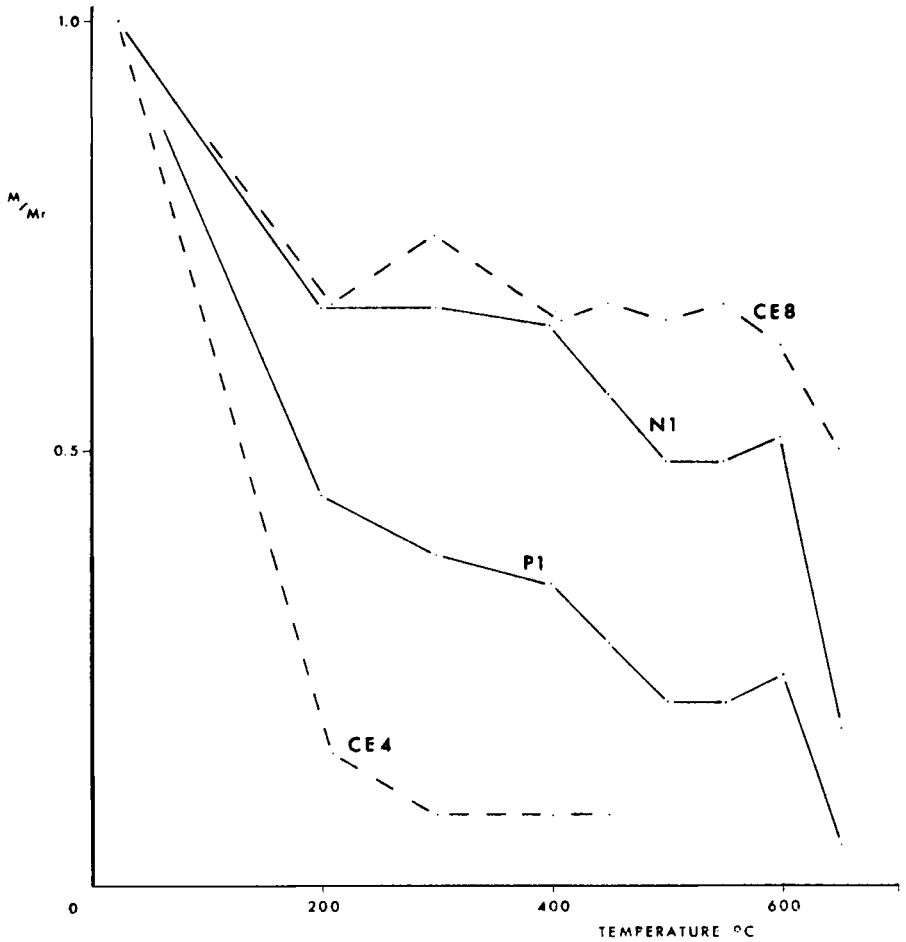


FIG. 4. Hudson Formation—thermal demagnetization curves. The curves illustrate the range of thermal characteristics for samples collected by Irving (dashed lines) and for the samples recently collected (solid lines).

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in that instance was rather conservative. In the case of the Hudson Formation, it is desirable to devise some formula for eliminating the results from samples that have been chemically remagnetized to an appreciable extent. A complication arises because of the similarities in the magnetic behaviour of all the recently collected samples.

Chemical remagnetization associated with weathering is a process common to many sediments of northern Australia (see earlier sections), and it is therefore not surprising that a large collection of rocks from this area should include some remagnetized samples. Nevertheless, reliable data can still be obtained, as in the case of the Jinduckin Formation where, although site E had obviously been magnetically altered, the removal of VRM from the remaining majority of samples produced stable directions of magnetization well away from the Earth's present field. In the case of the Hudson Formation, however, most of the samples have been chemically remagnetized and the remainder give directions much closer than the Jinduckin Formation to that of the present geomagnetic field. Further, the origin of the magnetization has not been substantiated by the collection of samples reversely magnetized with respect to the others.

When assessing data from the Hudson Formation, it is evident that differences in the behaviour of samples recently collected are most readily resolved in terms of the (streaked) cleaned directions of magnetization, since some remained close to the Earth's present field, some moved away to a quite distinct stable direction and some adopted intermediate positions.

Although some specimens were heated to temperatures as high as 650 °C in an effort to remove all secondary components, the optimum cleaning temperatures were 550–600 °C for the recently collected rocks and only 450 °C for Irivng's samples. The existence in most samples of a component stable at these temperatures is illustrated by three of the demagnetization curves in Fig. 4, which reveal 'plateaus' in the range 500–600 °C for Nl and Pl and a flatness above 200 °C for CE8. The curves delimit the range of thermal demagnetization characteristics for each collection, thus demonstrating that all of the more recently collected samples possessed curves having very similar shapes even though both primary and secondary stable magnetizations were probably represented. The presence of Curie temperatures exceeding 600 °C suggests that haematite is the dominant magnetic mineral in these samples. The curve for CE4 is typical of the weakly magnetized samples from Irving's site 1, three of which showed no within-sample agreement whilst the magnetic intensity of a fourth was so low that it approached the background 'noise' of the magnetometer.

A selection of representative specimens was subjected to a.f. demagnetization, the two samples from site R responding favourably. In the latter case, the NRM directions had suggested a possible within-site reversal, an issue that was not clarified by thermal cleaning. However, after magnetic washing, both samples disclosed stable directions (Fig. 5) similar to the accepted cleaned directions of Fig. 6, although that of R2 was much closer to the present geomagnetic field. It seems that the NRM at site R was predominantly an isothermal type, reflecting the influence of lightning strikes; this explanation is endorsed by the nature of the site's hillside exposure.

In comparison, most of Irving's samples possessed smaller components of secondary magnetization, and the cleaned (primary) directions, although dispersed, were further from the present geomagnetic field direction (Fig. 6). This could imply the possible non-removal of a substantial secondary magnetization from some or all of the samples recently collected, and it seems probable that the mean cleaned direction from Irving's samples is more likely to represent an original magnetization.

In the absence of any more definitive criteria arising from the laboratory measurements, it has been decided to reject all cleaned results lying within 30° of the Earth's present dipolar field. Many samples, including R2, are rejected on this principle. Sample M2 is rejected on the grounds that it was uniquely magnetized, as suggested

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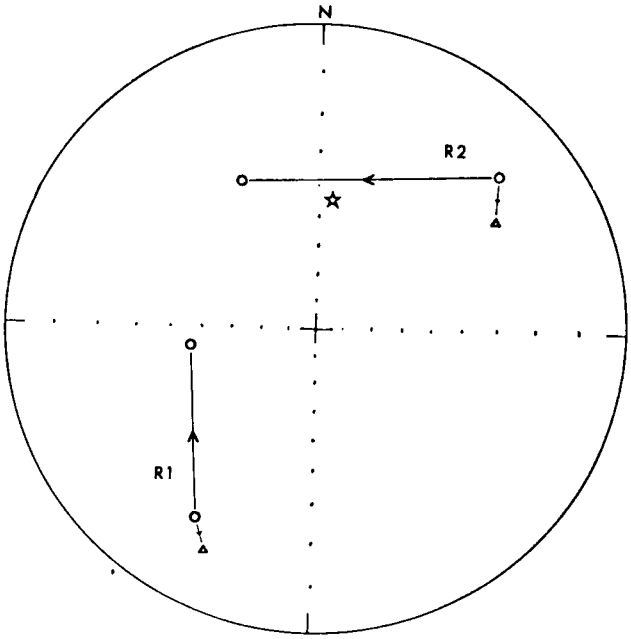


FIG. 5. Hudson Formation—directions of magnetization from site R. The circles show the substantial change in the mean direction of each sample after a.f. demagnetization. The triangles indicate directions after thermal treatment—showing very little movement from NRM values.

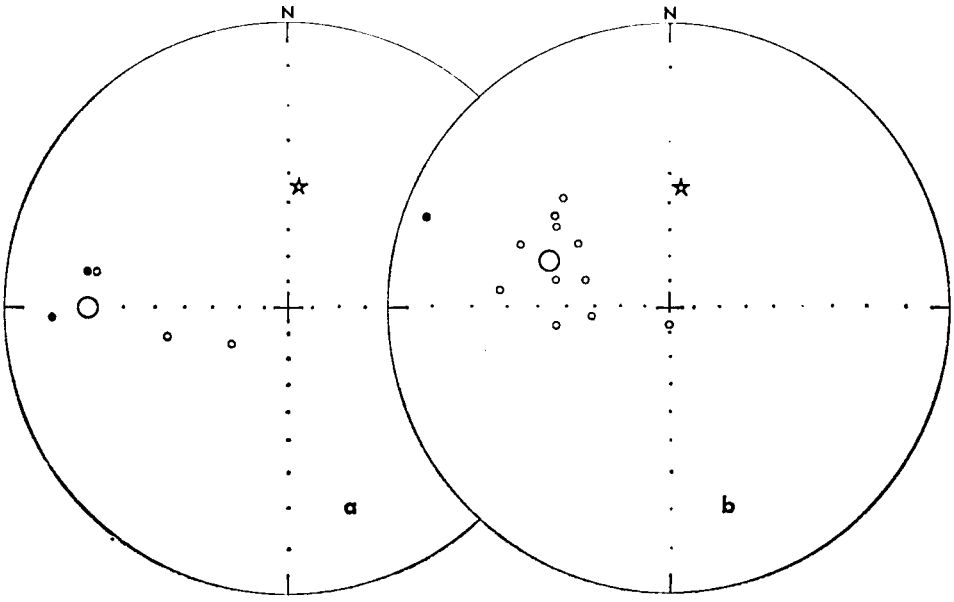


FIG. 6. Hudson Formation—cleaned directions. The directions of a supposed original magnetization from samples collected (*a*) by Irving, and (*b*) recently. The large circles indicate the mean direction for each collection. (The star represents the present geomagnetic field direction).

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by its NRM direction noted in Fig. 3(*b*). A total of six samples (including four collected by Irving) was rejected for having no within-sample agreement.

It is with some reservation that the formation mean direction of a supposed primary magnetization (Table 3) has been calculated from the sample results listed in Table 2. For this purpose all the accepted samples from Irving's collection (Fig. 6(*a*)) and the new collection (Fig. 6(*b*)) have been combined. The 17 samples give a palaeomagnetic pole position at 19° E, 18° N with $A_{95} = 13°$ but there is a strong possibility that this has been influenced by recent chemical magnetizations.

Table 3

Formation mean results after cleaning

		Mean direction					Mean of sample poles				
	$N(n)$	D	I	R	k	α_{95}	R	k	A_{95}	Long.	Lat.
					Hudson Formation						
Accepted data	17(56)	283	−37	14·7	7·0	14	15·0	8·1	13	19° E	18° N
Rejected data*	24(68)	348	−40	23·0	22·3	6					
					Jinduckin Formation						
Previously published	7(20)†	252	−14	6·38	9·6	20	6·57	14·0	17	29° E	16° S
Amended results	20(43)	252	−22	17·5	7·7	13	18·1	10·1	11	25° E	13° S

N = Number of hand samples (except in case marked †)

Other symbols and conventions as for Table 2

* Results that incorporate large component of present geomagnetic field, direction (004, −45)

†$N(n)$ represents numbers of sites (samples)

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Department of Geophysics & Geochemistry,
Australian National University,
Box 4, *P.O.,*
Canberra, A.C.T. 2601, *Australia.*

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